

Datasets Used

To evaluate the proposed implementation under different workload and complexity conditions, metagenomic datasets of different origins and sizes were used. The selection included public datasets widely used in the literature, reference data employed in previous metagenomic assembly studies, and microbial community sequences obtained from anaerobic sludge from a beverage industry facility in Uruguay. This diversity made it possible to validate the implementation, analyze its performance under different scenarios, and evaluate the scalability of the proposed acceleration strategy. Table 1 summarizes the main datasets used in the experimental evaluation.

Table 1. Datasets used in the experimental evaluation.

Acronym	Source	Number of reads	Technology	Use in this study
GAL_1026k	Galaxy Tutorial [27], [28]	1,026,109	Illumina	Reproducible validation
CAM_12739k	Campanaro et al. [9]	12,739,564	Illumina	Comparison with previous studies
AS_C1_22230k	Institute of Chemical Engineering (Fing-Udelar)	22,230,277	Illumina NovaSeq 6000	Main evaluation
AS_C2_20101k	Institute of Chemical Engineering (Fing-Udelar)	20,101,162	Illumina NovaSeq 6000	Main evaluation
AS_C3_3994k	Institute of Chemical Engineering (Fing-Udelar)	3,994,180	Illumina NovaSeq 6000	Main evaluation
AS_C4_5867k	Institute of Chemical Engineering (Fing-Udelar)	5,867,316	Illumina NovaSeq 6000	Main evaluation
AS_C5_6283k	Institute of Chemical Engineering (Fing-Udelar)	6,283,699	Illumina NovaSeq 6000	Main evaluation
AS_C6_7135k	Institute of Chemical Engineering (Fing-Udelar)	7,135,312	Illumina NovaSeq 6000	Main evaluation

The AS_C1-AS_C6 datasets correspond to microbial community samples from anaerobic sludge sequenced using Illumina NovaSeq 6000 technology. These datasets constituted the main evaluation set for the proposed implementation because of their representativeness and the large volume of data processed. The FASTQ files corresponding to each dataset are listed in Table 2. In every case, each dataset is composed of two files containing the *paired-end* reads used as input for assembly.

Table 2. FASTQ files corresponding to the datasets used.

Acronym	FASTQ files used
GAL_1026k	ERR2231568_1.fastqsanger.gz ERR2231568_2.fastqsanger.gz
CAM_12739k	SRR341725_1.fastq.gz SRR341725_2.fastq.gz
AS_C1_22230k	C1_FKDN220243764-1A_HN3V5DSX3_L3_1.fq.gz C1_FKDN220243764-1A_HN3V5DSX3_L3_2.fq.gz
AS_C2_20101k	C2_FKDN220243765-1A_HN3V5DSX3_L3_1.fq.gz C2_FKDN220243765-1A_HN3V5DSX3_L3_2.fq.gz
AS_C3_3994k	C3_FKDN220243766-1A_HN3V5DSX3_L3_1.fq.gz C3_FKDN220243766-1A_HN3V5DSX3_L3_2.fq.gz
AS_C4_5867k	C4_FKDN220243767-1A_HN3V5DSX3_L3_1.fq.gz C4_FKDN220243767-1A_HN3V5DSX3_L3_2.fq.gz
AS_C5_6283k	C5_FKDN220243768-1A_HN3V5DSX3_L3_1.fq.gz C5_FKDN220243768-1A_HN3V5DSX3_L3_2.fq.gz
AS_C6_7135k	C6_FKDN220243769-1A_HN3V5DSX3_L3_1.fq.gz C6_FKDN220243769-1A_HN3V5DSX3_L3_2.fq.gz

In addition to the datasets described above, increasingly large subsets were generated by subsampling the AS_C1_22230k dataset while preserving the correspondence between the FASTQ files and, therefore, their *paired-end* structure. The sizes considered were 5k, 10k, 50k, 70k, 100k, 150k, 200k, 250k, 300k, and 350k read pairs, where k denotes one thousand pairs. These subsets were used during implementation development and debugging and to analyze how the execution time of the GPU-accelerated routines evolved as the number of processed read pairs increased.

Experiments conducted on these subsets were used to analyze the scalability of the implementation and characterize the behavior of the Unitig Graph construction routines as the problem size increased. These datasets also supported functional validation and incremental code debugging throughout the different stages of development.

The datasets and experimental platform described in the preceding sections formed the basis for evaluating the proposed implementation. The following section describes the methodology used to compare assembly performance and quality between the CPU and GPU versions.